

STAT 4005 - Time Series

Chapter 2 - Probability Models

Agenda

1 Introduction

2 Stochastic process

- Definition 1 – Stochastic process
- Definition 2 – Finite dimensional distribution function
- Definition 3 – Strictly stationary
- Definition 4 – Weakly stationary
- Definition 5 – Autocovariance function

3 Strictly vs weakly stationary

4 Examples

5 Sample correlation function

6 Summary

- Assume that we have done the preliminary analysis to estimate trend T_t and seasonality S_t
- What remains unexplained is the noise N_t :

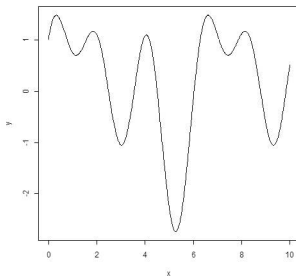
$$N_t = X_t - T_t - S_t$$

- We propose statistical models to describe the noise $\{N_t\}$ to extract more information

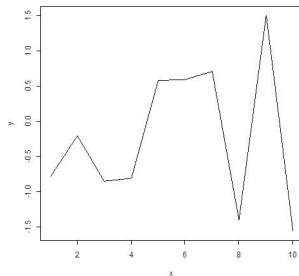
Definition 1 – Stochastic process

- A collection of random variable $\{X_t : t \in \mathcal{T}\}$ is called a stochastic process, where $\mathcal{T}$ is an index set of time.
- Continuous
 $\{X(t) : t \in [0, \infty)\}$
- Discrete
 $\{X_t : t \in \{1, 2, \dots, n\}\}$

Continuous Time Process



Discrete Time Process



Definition 1 – Stochastic process

Understand the meaning of **stochastic** (Random):

- $\{X_t : t = 1, 2, \dots, n\}$ can be regarded as a function of time t
- We assume that there is a sample space Ω (a set of all possible scenario) that **generates** the randomness
 - $\Omega = (\omega_1, \omega_2, \omega_3, \dots)$ (an ∞ -face die)
 $\Rightarrow \omega_1 \rightarrow \{X_t(\omega_1)\}; \omega_2 \rightarrow \{X_t(\omega_2)\}$ (each face gives one path)
- For a fixed ω , $\{X_t(\omega)\}$ is called a sample function/ realization/ sample path
- For a fixed t , $X_t(\cdot)$ is a random variable
- In practice, we can only observe one sample path (i.e. the situation is generated by one particular ω)
- To summarize important features from the data, we require the process to be repeating itself in some way (stationarity).

Definition 2 – Finite dimensional distribution function

- Finite dimensional distribution functions:

$$F_{\mathbf{t}}(\mathbf{x}) = P(X_{t_1} \leq x_1, \dots, X_{t_n} \leq x_n)$$

for all possible $\mathbf{t} = (t_1, t_2, \dots, t_n)$ where, $1 \leq t_1 < t_2 < \dots < t_n \leq \infty$ and $\mathbf{x} = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n, n = 1, 2, \dots$

Definition 3 – Strictly stationary

- Strictly stationary - distribution of a process does not change over time.
- $\{X_t\}$ is said to be strictly stationary if
 - for all n ,
 - for all $(t_1, t_2, \dots, t_n)$,
 - for all h ,

$$(X_{t_1}, \dots, X_{t_n}) \stackrel{d}{=} (X_{t_1+h}, \dots, X_{t_n+h}) ,$$

where " $\stackrel{d}{=}$ " means "equal in distribution", i.e.,

$$F_{\mathbf{t}}(\mathbf{x}) = P(X_{t_1} \leq x_1, \dots, X_{t_n} \leq x_n) = P(X_{t_1+h} \leq x_1, \dots, X_{t_n+h} \leq x_n) = F_{\mathbf{t}+h}(\mathbf{x})$$

- May not be easy to check in practice because joint distribution $F_{\mathbf{t}}(\mathbf{x})$ is difficult to compute.

Definition 4 – Weakly stationary

- Weakly Stationary:

- same mean
- same variance and same covariance

across time.

- Formally, $\{X_t\}$ is weakly stationary/ second order stationary/ wide-sense stationary if

- 1) $E(X_t) = \mu$
- 2) $\text{Cov}(X_t, X_{t+h}) = \gamma(h)$ for all t and h .
(auto-covariance only depends on time lag h , but not time t)

- Easier to verify weakly stationary than strictly stationary

Definition 5 – Autocovariance function

- Autocovariance function (ACVF)

$$\gamma(h) = \text{Cov}(X_t, X_{t+h})$$

- Autocorrelation function (ACF)

$$\begin{aligned}\rho(h) &= \frac{\text{Cov}(X_t, X_{t+h})}{\sqrt{\text{Var}(X_t)\text{Var}(X_{t+h})}} \\ &= \frac{\gamma(h)}{\gamma(0)}.\end{aligned}$$

Under stationarity, we have the following properties:

- $\gamma(0) = \text{Var}(X_t) = \text{Var}(X_{t+h})$
- $\gamma(-h) = \gamma(h)$.
- $\rho(-h) = \rho(h)$.

Exercise

Let $X_t = (-1)^t Z$, where Z is a random variable. Give necessary condition(s) on Z so that X_t is weakly stationary.

Solution:

$$\begin{aligned} E(X_t) &= (-1)^t E(Z), \\ \text{Cov}(X_t, X_{t+k}) &= (-1)^{2t+k} \text{Cov}(Z, Z) = (-1)^k \text{Var}(Z). \end{aligned}$$

Note that the autocovariance function is independent of t . To make the mean constant we need $(-1)^t E(Z)$ to be independent of t , which implies $E(Z) = -E(Z)$, i.e. $E(Z) = 0$. Therefore, we require $E(Z) = 0$ if $\{X_t\}$ is stationary.

Strictly vs weakly stationary

- Strictly: distribution of the process unchanged.
Weakly: mean/covariance unchanged.
- Strictly stationary implies weakly stationary whenever $E(X_t^2) < \infty$.
- Weakly stationary implies strictly stationary when $\{X_t\}$ is normal, but in general not true.
- In the subsequent chapters, unless indicated specifically, stationary always refers to weakly stationary.

Strictly vs weakly stationary

- Basic statistics: If $X \sim t_{df}$, then $E(X^k) < \infty$ if and only if $k < df$.
1. $X_1, X_2, \dots, X_n \stackrel{\text{iid}}{\sim} t_2$
 - $\{X_t\}$ is strictly stationary
 - But $E(X_t^2) = \infty$,
 $\therefore \{X_t\}$ is **NOT** weakly stationary
 2. $X_1, X_2, \dots, X_n \stackrel{\text{iid}}{\sim} t_3$
 - Now $E(X_t^2) < \infty$,
 $\therefore \{X_t\}$ is strictly and weakly stationary
 3. $X_1 \sim \text{Exp}(1) - 1, X_2, X_3, \dots \stackrel{\text{iid}}{\sim} N(0, 1)$
 - $\{X_t\}$ is weakly stationary:
 $E(X_t) = 0, \text{Var}(X_t) = 1, \text{Cov}(X_t, X_{t+k}) = 0$, all t and $k \neq 0$
 - But $\{X_t\}$ is **NOT** strictly stationary, since $X_1 \stackrel{d}{\neq} X_2$.

Exercise

Let $W_t = (-1)^t Y_t$ where Y_t are independently and identically distributed with $P(Y_t = k) = 0.2, k = -2, -1, 0, 1, 2$ for all integers t .

- (a) Is W_t strictly stationary?
- (b) Is W_t weakly stationary?

Solution:

- (a) Note that $P(-Y_t = k) = 0.2, k = -2, -1, 0, 1, 2$ for all integers t . Thus Y_t and $-Y_t$ have the same distribution. Since $\{Y_t\}_{t=1,\dots}$ are independent, the series $\{(-1)^t Y_t\}_{t=1,\dots}$ are actually independent and identically distributed. Thus, $\{W_t\}$ is strictly stationary.
- (b) Since

$$E(W_t^2) = E(Y_t^2) = 0.2((-2)^2 + (-1)^2 + 0^2 + 1^2 + 2^2) = 2 < \infty,$$

the second moment of W_t exists. Thus the strict stationarity of $\{W_t\}$ implies its weak stationarity.

Exercise

Give an example of a series $\{Z_t\}$ with a constant mean and $\text{Corr}(Z_t, Z_{t+k})$ being free of t , but $\{Z_t\}$ is not stationary.

Solution: Consider $Z_t = ta_t$, where $a_t \stackrel{\text{iid}}{\sim} N(0, 1)$. By straightforward algebra,

$$\begin{aligned} E(Z_t) &= tE(a_t) = 0, \\ \text{Var}(Z_t) &= t^2\text{Var}(a_t) = t^2, \\ \text{Cov}(Z_t, Z_{t+k}) &= t(t+k)\text{Cov}(a_t, a_{t+k}) = 0, \quad \text{for } k \neq 0, \\ \text{Corr}(Z_t, Z_{t+k}) &= \frac{t(t+k)\text{Cov}(a_t, a_{t+k})}{\sqrt{t^2(t+k)^2}} = \begin{cases} 1, & \text{for } k = 0, \\ 0, & \text{for } k \neq 0, \end{cases} \end{aligned}$$

Clearly, $E(Z_t)$ and $\text{Corr}(Z_t, Z_{t+k})$ are independent of time t . But $\{Z_t\}$ is not stationary since the variance $\text{Var}(Z_t) = t^2$ depends on time.

Example 1: White Noise

If a time series $\{X_t\}$ is a sequence of uncorrelated random variables with

$$\rho(h) = \begin{cases} 1, & h = 0, \\ 0, & |h| \geq 1, \end{cases} \quad (1)$$

with zero mean and constant variance, then it is called a **white noise** sequence.

- i.i.d. random variables are white noise if their second moments exist
- white noise *may not be* i.i.d.
(e.g., $X_1 \sim \text{Exp}(1) - 1$, $X_k \sim N(0, 1)$, $k \geq 2$)
- independence is a stronger assumption than zero-correlation

Exercise

Show that $Z_t = a_t a_{t-2}$ is a white noise process, where a_t are i.i.d. sequence of random variables with zero mean and variance σ_a^2 .

Solution: Note that $E(Z_t) = E(a_t a_{t-2}) = 0$. In addition, for $k \neq 0$,

$$\begin{aligned}\text{Cov}(Z_t, Z_{t+k}) &= E(Z_t Z_{t+k}) - E(Z_t)E(Z_{t+k}) \\ &= E(a_t a_{t-2} a_{t+k} a_{t+k-2}) - 0 \\ &= 0 \\ \text{Var}(Z_t) &= E(Z_t^2) \\ &= E(a_t^2)E(a_{t-2}^2) \\ &= \sigma_a^4 < \infty.\end{aligned}$$

Thus, the mean of $\{Z_t\}$ is constant, and the autocorrelation function is the same as that of a white noise process, so $Z_t \sim WN(0, \sigma_a^4)$.

Example 2: Same Observation

If $\{Y_t\}$ satisfies $Y_1 = Y_2 = \dots = Y$ with $E(Y^2) < \infty$, then

- $\rho(h) = \text{Corr}(Y, Y) = 1$ for all h .
- Essentially there is only one observation.
- In comparison with White Noise in Example 1, we have

$$\begin{aligned}\frac{1}{n}(X_1 + \dots + X_n) &\rightarrow E(X_1) = \mu, \\ \frac{1}{n}(Y_1 + \dots + Y_n) &= Y.\end{aligned}$$

There is as much randomness in n -th sample average as there is in the first observation Y_1 .

Example 3: Periodic Function

- Suppose a time series $\{X_t\}$ satisfies

$$X_t = A \cos \theta t + B \sin \theta t,$$

where A and B are i.i.d. random variables with mean 0 and variance σ^2 .

- $\{X_t\}$ is weakly stationary because
 - $E(X_t) = E(A) \cos \theta t + E(B) \sin \theta t = 0$
 -

$$\begin{aligned} & \text{Cov}(X_{t+h}, X_t) \\ &= \text{Cov}(A \cos \theta(t+h) + B \sin \theta(t+h), A \cos \theta t + B \sin \theta t) \\ &= \cos \theta t \cos \theta(t+h) \text{Var}(A) + \sin \theta t \sin \theta(t+h) \text{Var}(B) \\ &= \sigma^2 \cos(\theta(t+h) - \theta t) \quad (\cos(A-B) = \cos A \cos B + \sin A \sin B) \\ &= \sigma^2 \cos \theta h, \end{aligned}$$

independent of t .

Example 4: Moving Average MA(1) Model

Let $\{\epsilon_t\}$ be a white noise sequence with $\text{Var}(\epsilon_t) = 1$ and $X_t = \epsilon_t + 0.5\epsilon_{t-1}$. Is $\{X_t\}$ stationary? Find the ACVF and ACF of $\{X_t\}$.

$$E(X_t) = 0.$$

$$\gamma(h) = \begin{cases} 1.25, & h = 0, \\ 0.5, & h = 1, \\ 0, & h \geq 2. \end{cases}$$

$$\rho(h) = \begin{cases} 1, & h = 0, \\ 0.4, & h = 1, \\ 0, & h \geq 2. \end{cases}$$

Since $E(X_t)$ and $\rho(h)$ are independent of t , $\{X_t\}$ is weakly stationary.

Exercise

Suppose that $\{a_t\}_{t=1,\dots}$ are independent and identically distributed random variables with mean 0 and finite variance σ_a^2 , and θ_i are constants for $i = 0, 1, 2, 3, 4$. If

$$X_t = -\theta_0 a_t - \theta_1 a_{t-1} - \theta_2 a_{t-2} - \theta_3 a_{t-3} - \theta_4 a_{t-4},$$

then compute the autocorrelation function ρ_k of X_t for $k = 0, 1, 2, \dots$

Solution:

$$\text{Var}(X_t) = \text{Var}(-\theta_0 a_t - \theta_1 a_{t-1} - \theta_2 a_{t-2} - \theta_3 a_{t-3} - \theta_4 a_{t-4})$$

$$= (\theta_0^2 + \theta_1^2 + \theta_2^2 + \theta_3^2 + \theta_4^2) \sigma_a^2$$

$$\text{Cov}(X_t, X_{t+1}) = (\theta_0 \theta_1 + \theta_1 \theta_2 + \theta_2 \theta_3 + \theta_3 \theta_4) \sigma_a^2$$

$$\text{Cov}(X_t, X_{t+2}) = (\theta_0 \theta_2 + \theta_1 \theta_3 + \theta_2 \theta_4) \sigma_a^2$$

$$\text{Cov}(X_t, X_{t+3}) = (\theta_0 \theta_3 + \theta_1 \theta_4) \sigma_a^2$$

$$\text{Cov}(X_t, X_{t+4}) = (\theta_0 \theta_4) \sigma_a^2.$$

Exercise (Continued)

Thus, the autocorrelation function is

$$\begin{aligned}\rho(0) &= 1 \\ \rho(1) &= \frac{\theta_0\theta_1 + \theta_1\theta_2 + \theta_2\theta_3 + \theta_3\theta_4}{\theta_0^2 + \theta_1^2 + \theta_2^2 + \theta_3^2 + \theta_4^2} \\ \rho(2) &= \frac{\theta_0\theta_2 + \theta_1\theta_3 + \theta_2\theta_4}{\theta_0^2 + \theta_1^2 + \theta_2^2 + \theta_3^2 + \theta_4^2} \\ \rho(3) &= \frac{\theta_0\theta_3 + \theta_1\theta_4}{\theta_0^2 + \theta_1^2 + \theta_2^2 + \theta_3^2 + \theta_4^2} \\ \rho(4) &= \frac{\theta_0\theta_4}{\theta_0^2 + \theta_1^2 + \theta_2^2 + \theta_3^2 + \theta_4^2} \\ \rho(k) &= 0, \quad |k| > 4\end{aligned}$$

Exercise

Consider the process $Z_t = a_t - 1.7a_{t-1} + 0.72a_{t-2}$ where $\{a_t\}$ are i.i.d. with finite variance $\sigma_a^2 = 4$.

- (1) Find $\text{Var}(Z_t)$.
- (2) Find the autocovariance function of the process.
- (3) Find $\text{Var}(\frac{1}{15} \sum_{t=1}^{15} Z_t)$.

Solution

(1)

$$\begin{aligned}\text{Var}(Z_t) &= \text{Var}(a_t - 1.7a_{t-1} + 0.72a_{t-2}) \\ &= (1 + (1.7)^2 + (0.72)^2)\sigma_a^2 = 17.6\end{aligned}$$

(2)

$$\begin{aligned}\gamma(0) &= \text{Var}(Z_t) = 17.6 \\ \gamma(1) &= \text{Cov}(Z_t, Z_{t+1}) = -11.7 \\ \gamma(2) &= \text{Cov}(Z_t, Z_{t+2}) = 2.88 \\ \gamma(k) &= 0 \quad |k| > 2\end{aligned}$$

(3)

$$\begin{aligned}\text{Var}\left(\frac{1}{15}\sum_{t=1}^{15} Z_t\right) &= \frac{1}{15}\gamma(0) + \frac{1}{15^2}2(15-1)\gamma(1) + \frac{1}{15^2}2(15-2)\gamma(2) \\ &= 0.05.\end{aligned}$$

Example 5: Moving Average $MA(\infty)$ Model

Let $\{\epsilon_t\}$ be a white noise sequence with $\text{Var}(\epsilon_t) = 1$ and $X_t = \sum_{k=0}^{\infty} \psi_k \epsilon_{t-k}$ and $\sum_{k=0}^{\infty} \psi_k^2 < \infty$. Is $\{X_t\}$ stationary? Find the ACVF and ACF of $\{X_t\}$.

$$E(X_t) = 0.$$

$$\gamma(h) = \begin{cases} \sum_{k=0}^{\infty} \psi_k^2, & h = 0, \\ \sum_{k=0}^{\infty} \psi_k \psi_{k+h}, & h \geq 1. \end{cases}$$

$$\rho(h) = \begin{cases} 1, & h = 0, \\ \frac{\sum_{k=0}^{\infty} \psi_k \psi_{k+h}}{\sum_{k=0}^{\infty} \psi_k^2}, & h \geq 1. \end{cases}$$

Note that by Cauchy-Schwarz inequality,

$$\sum_{k=0}^{\infty} |\psi_k \psi_{k+h}| \leq \sqrt{\left(\sum_{k=0}^{\infty} \psi_k^2\right) \left(\sum_{k=0}^{\infty} \psi_{k+h}^2\right)} < \infty,$$

hence the sum $\sum_{k=0}^{\infty} \psi_k \psi_{k+h}$ is well defined. Since $E(X_t)$ and $\rho(h)$ are independent of t , $\{X_t\}$ is weakly stationary.

For the $MA(\infty)$ model to be well defined, we usually require that $\sum_{k=0}^{\infty} |\psi_k| < \infty$. Such that the sum

$$X_t = \sum_{k=0}^{\infty} \psi_k \epsilon_{t-k}$$

converges absolutely. This is a stronger condition because

$$\sum_{k=0}^{\infty} \psi_k^2 \leq \left(\sum_{k=0}^{\infty} |\psi_k| \right)^2 < \infty.$$

Sample ACVF and ACF

$$\text{ACVF} : \gamma(h) = \text{Cov}(X_t, X_{t+h})$$

$$\text{ACF} : \rho(h) = \frac{\text{Cov}(X_t, X_{t+h})}{\sqrt{\text{Var}(X_t)\text{Var}(X_{t+h})}} = \frac{\gamma(h)}{\gamma(0)}$$

- Both ACVF and ACF are *population quantities*. It means that they involve the unknown true distribution of the process $\{X_t\}$
- If we have a dataset $X_1, X_2, \dots, X_n$ but we do not know the true probability distribution of the time series, we can estimate the ACF/ACVF by
 - Sample ACVF: $C_h = \frac{1}{n} \sum_{t=1}^{n-h} (X_t - \bar{X})(X_{t+h} - \bar{X})$
 - Sample ACF : $r_h = C_h / C_0$

Sample ACVF and ACF

- Sample ACVF: $C_h = \frac{1}{n} \sum_{t=1}^{n-h} (X_t - \bar{X})(X_{t+h} - \bar{X})$
- Why $\frac{1}{n}$ but not $\frac{1}{n-h}$?

- To ensure the **non-negative definiteness** of covariance matrix

$$\widehat{\text{Var}}((X_1, \dots, X_n)) = (C_{|i-j|})_{i,j \in \{1, \dots, n\}} = \frac{1}{n} M M',$$

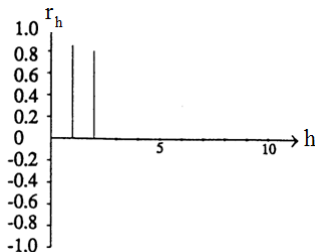
where with $\tilde{X}_k = X_k - \bar{X}$,

$$M = \begin{pmatrix} \tilde{X}_1 & \tilde{X}_2 & \tilde{X}_3 & \cdots & \tilde{X}_n & 0 & 0 & 0 \\ 0 & \tilde{X}_1 & \tilde{X}_2 & & \tilde{X}_{n-1} & \tilde{X}_n & 0 & 0 \\ 0 & 0 & \tilde{X}_1 & & \tilde{X}_{n-2} & \tilde{X}_{n-1} & \tilde{X}_n & 0 \\ & & & \ddots & & & & \\ & & & & \tilde{X}_1 & \tilde{X}_2 & & \tilde{X}_n \end{pmatrix}.$$

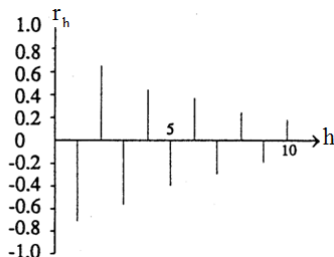
- Denote $\mathbf{X} = (X_1, \dots, X_n)$,
 $\widehat{\text{Var}}(\mathbf{a}'\mathbf{X}) = \mathbf{a}'\widehat{\text{Var}}(\mathbf{X})\mathbf{a} = \frac{1}{n}(\mathbf{a}'M)(\mathbf{a}'M)' \geq 0$ implies that
 $(C_{|i-j|})_{i,j \in \{1, \dots, n\}}$ is non-negative definite.

Correlogram - ACF plot

Correlogram - ACF plot: A graph to summarize r_h (`acf(x)` in R)



Example 1

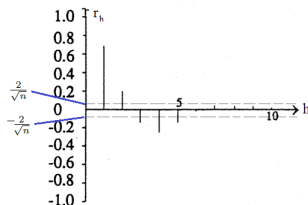


Example 2

- Trust r_h up to $h = n/3$, otherwise r_h is not accurate
- By definition, $r_0 = \frac{C_0}{C_0} = 1$

Sample Correlation Function

- Questions commonly of interest:
 - Is $\text{Corr}(X_t, X_{t+h}) = 0$?
 - ⇔ Is r_h significantly different from 0 ?
- Solution:
 - Results: If $\{X_t\}$ is white noise, then $r_h \sim N(0, \frac{1}{n})$ for each $h \geq 1$ when n is large.
 - Rigorous proof is out of the scope of this course, interested students may refer to Chapter 6 of the following book.
 - Brockwell, P. J., and Davis, R. A. (1991). Time series: theory and methods. Springer.
 - Draw horizontal lines at levels $\pm \frac{2}{\sqrt{n}}$ on acf and check for exceedance.



Example: ACF plot

Scratch the sample ACF plot for the dataset $(2, 5, 3, 5, 6, 3, 4)$. Is this time series a white noise sequence?

Since $n = 7$, the maximum value of h is the integer part of $7/3$ or 2. We have

$$\overline{X} = 4,$$

$$C_0 = \frac{1}{7} \sum_{t=1}^7 (X_t - \overline{X})^2 = 1.714286,$$

$$C_1 = \frac{1}{7} \sum_{t=1}^6 (X_t - \overline{X})(X_{t+1} - \overline{X}) = -0.5714286,$$

$$C_2 = \frac{1}{7} \sum_{t=1}^5 (X_t - \overline{X})(X_{t+2} - \overline{X}) = 0,$$

$$r_0 = 1,$$

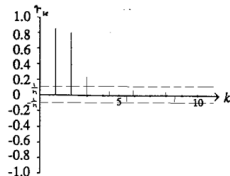
$$r_1 = \frac{C_1}{C_0} = -0.333333,$$

$$r_2 = 0.$$

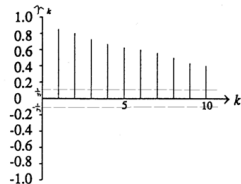
Since $\frac{2}{\sqrt{n}} = 0.7559289$, all the sample ACF are insignificant. We may conclude that the time series is a white noise sequence.

Short memory vs long memory

- Short Memory = Short term correlation

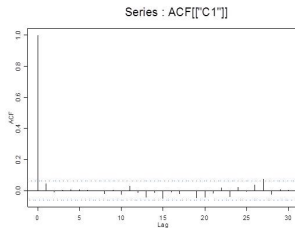


- Long Memory = long term correlation

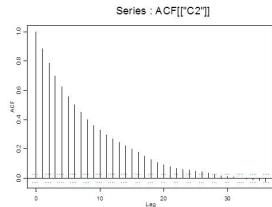


More examples of ACF function

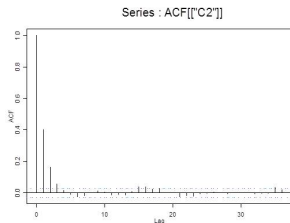
- $Y_t = Z_t, \{Z_t\}$ are i.i.d.



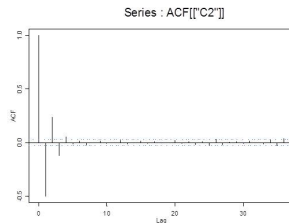
- $Y_t = 0.9Y_{t-1} + Z_t$



- $Y_t = 0.4Y_{t-1} + Z_t$

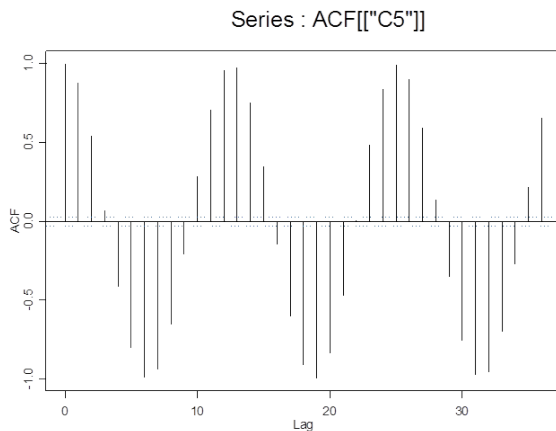


- $Y_t = -0.5Y_{t-1} + Z_t$



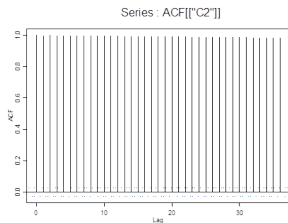
More examples of ACF function

- $Y_t = a \cos(t\omega)$, where a, ω are constants

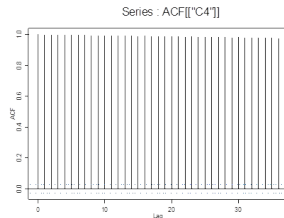


More examples of ACF function

- $Y_t = Y_{t-1} + Z_t$



- $Y_t = at + Z_t$



1. In both cases, ACF decay very slowly
2. Commonly observed in **non-stationary** process
3. In practice, perform detrending/filtering and study the resulting stationary noises

Remarks on sample ACF function

- Notice that the examples above suggest that we can "identify" a time series through inspection of its ACF.
- Although this sounds promising, it is not a procedure that is always free of error.
- When we calculate the ACF of any given series with a fixed sample size n , we cannot put too much confidence in the values of r_k for large value of k .
- Furthermore, if there is a trend in the data,

$$X_t = T_t + N_t.$$

then X_t becomes nonstationary and the idea of inspecting the ACF of X_t becomes questionable. It is therefore important to detrend the data before interpreting their ACF.

Summary of Chapter 2

- Strictly Stationary: Joint distribution remain unchanged
- Weakly Stationary: Mean and covariance structure remain unchanged
- White Noise: $E(X_t) = 0$, $\text{Cov}(X_t, X_k) = \sigma^2 1_{\{t=k\}}$

$$\text{ACVF} : \gamma(h) = \text{Cov}(X_t, X_{t+h})$$

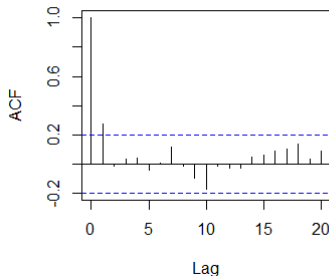
$$\text{ACF} : \rho(h) = \frac{\text{Cov}(X_t, X_{t+h})}{\sqrt{\text{Var}(X_t)\text{Var}(X_{t+h})}} = \frac{\gamma(h)}{\gamma(0)}$$

$$\text{Sample ACVF} : C_h = \frac{1}{n} \sum_{t=1}^{n-h} (X_t - \bar{X})(X_{t+h} - \bar{X})$$

$$\text{Sample ACF} : r_h = C_h / C_0$$

Summary of Chapter 2

ACF plot (ACF r_h against lag h):

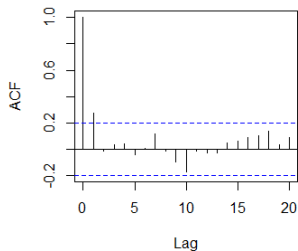


Check if $\rho(h) = \text{Corr}(X_t, X_{t+h}) = 0$:

- $\{X_t\}$ is white noise $\Rightarrow r_h \sim N(0, \frac{1}{n})$ for all $h \geq 1$
- Draw horizontal lines at levels $\pm \frac{2}{\sqrt{n}}$ on acf and check for exceedance.

Summary of Chapter 2

- Stationary



- Non-stationary

